$\boxed{2\text{-}6:}$ Reduce the following Boolean expressions to the required number of literals.

(a) $ABC+A'B'C+A'BC+ABC'+A'B'C'$ to five literals (Theorem 4(a))

$=((ABC+A'B'C)+A'BC)+(ABC'+A'B'C')$ (Postulate 3(a))

$=((CAB+CA'B')+A'BC)+(C'AB+C'A'B')$ (Postulate 4(a))

$=(C\cdot(AB+A'B')+A'BC)+(C'(AB+A'B'))$ (Postulate 3(a))

$=(C'(AB+A'B'))+(C\cdot(AB+A'B')+A'BC)$ (Theorem 4(a))

$=(C'(AB+A'B')+C\cdot(AB+A'B'))+A'BC$ (Postulate 3(b))

$=((AB+A'B')C'+(AB+A'B')\cdot C)+A'BC$ (Postulate 4(a))

$=(AB+A'B')(C'+C)+A'BC$ (Postulate 3(a))

~~= (AB+A'B')· [illegible]~~ $=(AB+A'B')(C+C')+A'BC$ (Postulate 3(a))

$=(AB+A'B')\cdot 1+A'BC$ (Postulate 5(a))

$=(AB+A'B')+A'BC$ (Postulate 2(b)) $=(A'B'+AB)+A'BC$ (Postulate 3(a))

$=A'B'+(AB+A'BC)$ (Theorem 4(a)) $=A'B'+(BA+BA'C)$ (Postulate 3(b))

$=A'B'+(B\cdot(A+A'C))$ (Postulate 4(a)) $=A'B'+(B\cdot((A+A')\cdot(A+C)))$ (Postulate 4(b))

$=A'B'+(B\cdot(1\cdot(A+C)))$ (Postulate 5(a)) $=A'B'+B\cdot((A+C)\cdot 1)$ (Postulate 3(b))

$=A'B'+B\cdot(A+C)$ (Postulate 2(b))

(b) $BC+AC'+AB+BCD$ to four literals

$=BCD+(BC+AC'+AB)$ (Postulate 3(a))

$=(BCD+BC)+(AC'+AB)$ (Theorem 4(a))

$=(BCD+BC\cdot 1)+(AC'+AB)$ (Postulate 2(b))

$=(BC\cdot(D+1))+(AC'+AB)$ (Postulate 4(a))

$=(BC\cdot 1)+(AC'+AB)$ (Theorem 2(a))

$=BC+(AC'+AB)$ (Postulate 2(b))

$=BC+(AB+AC')$ (Postulate 3(a))

$=(BC+AB)+AC'$ (Theorem 4(a))

$=(BC+BA)+AC'$ (Postulate 3(b))

~~$=B\cdot(C+A)+AC'$ (Postulate 4(a))~~